

A signed-gap decision criterion for accurate polynomial evaluation

Abstract

A finite criterion is proposed for the existence of accurate evaluation algorithms for real polynomials in the constant-free, independently rounded arithmetic model. For each ordering of the input magnitudes and each choice of their signs, write the inputs as signed cumulative sums of nonnegative gaps. Expand the given polynomial in these gaps. The criterion requires that the polynomial obtained by taking the absolute values of its coefficients be bounded by a constant times the absolute value of the polynomial itself, on the entire nonnegative orthant. The proof below establishes necessity by constructing a separated execution of an arbitrary finite branching program, then replaying it under multiplicative gap perturbations while holding every rounded value fixed. Polynomial interpolation yields coefficient domination. Sufficiency follows from direct evaluation in gap coordinates. The criterion is a finite collection of first-order sentences over the reals, so real quantifier elimination gives an always-halting decision procedure for AA-01. Stored values, exact negation, equality branches, nonhomogeneous polynomials, and boundary points are included.

Status. This is a proposed general solution, not a finite-case result. The argument and its implementation checks have not been independently reviewed or formally verified. No assertion of publication priority is made.

1 Problem and criterion

The target is AA-01 in the arithmetic-and-complexity section of *OpenProblemsInNLA*.¹ The underlying accurate-evaluation question and the independent-error model were developed by Demmel, Dumitriu and Holtz,² and surveyed with Koev.³ The result here concerns the precise finite-tree, whole-real-space version specified next, not every variation of the broader question.

Let $n \geq 1$ and $p \in \mathbb{Z}[x_1, \dots, x_n]$ with $p(0) = 0$. Inputs are exact real numbers. A finite computation tree uses rounded binary operations

$$(a, b) \mapsto (a + b)(1 + \delta_j), \quad (a - b)(1 + \delta_j), \quad ab(1 + \delta_j).$$

Each arithmetic occurrence has its own arbitrary error. Exact negation, comparisons of available real values, branching, and stored-value reuse are permitted. No real constants are input nodes. Comparison outcomes control branches; they are not additional numeric inputs. An accurate tree P must satisfy $P(x, 0) = p(x)$ and

$$\forall \eta \in (0, 1) \exists u \in (0, 1) \forall x \in \mathbb{R}^n \forall \delta \in [-u, u]^{N(P)} : \quad |P(x, \delta) - p(x)| \leq \eta |p(x)|. \quad (1)$$

Here $N(P)$ counts the rounded nodes in the finite control-flow tree; unused errors do not matter. The tree is fixed as η varies.

¹OpenProblemsInNLA, “AA-01—Deciding accurate evaluability of real polynomials,” statement checked 11 September 2026; repository statement.

²J. Demmel, I. Dumitriu and O. Holtz, *Toward accurate polynomial evaluation in rounded arithmetic*, arXiv:math/0508350v2, 18 January 2006, especially Sections 2.2–2.7 and 4.3; <https://arxiv.org/abs/math/0508350v2>.

³J. Demmel, I. Dumitriu, O. Holtz and P. Koev, *Accurate and efficient expression evaluation and linear algebra*, Acta Numerica 17 (2008), 87–145, especially Section 3.3.7; <https://doi.org/10.1017/S0962492906350015>.

1.1 Signed-gap charts

For a permutation $\pi \in S_n$ and signs $\sigma = (\sigma_1, \dots, \sigma_n) \in \{-1, 1\}^n$, define the invertible integer linear map $L = L_{\pi, \sigma}$ by

$$(Ly)_{\pi(i)} = \sigma_i \sum_{j=1}^i y_j, \quad i = 1, \dots, n. \quad (2)$$

The image of $\mathbb{R}_{\geq 0}^n$ is a closed signed-order chamber. These $2^n n!$ images cover $\mathbb{R}^n$. For an input in this chamber,

$$y_1 = |x_{\pi(1)}|, \quad y_i = |x_{\pi(i)}| - |x_{\pi(i-1)}| \quad (i > 1). \quad (3)$$

Chambers may overlap at zero inputs or tied magnitudes. No generic-position assumption is made.

For a polynomial $q(y) = \sum_{\alpha} c_{\alpha} y^{\alpha}$, always with collected coefficients, put

$$q^{\#}(y) = \sum_{\alpha} |c_{\alpha}| y^{\alpha}. \quad (4)$$

In particular, $q^{\#} \geq 0$ on the nonnegative orthant.

Theorem 1.1 (Signed-gap criterion). *For the model above, the following statements are equivalent.*

- (i) *An accurate finite computation tree for p exists.*
- (ii) *For every signed-gap chart L , the polynomial $q = p \circ L$ satisfies*

$$\exists C \geq 1 \ \forall y \in \mathbb{R}_{\geq 0}^n : \quad q^{\#}(y) \leq C |q(y)|. \quad (5)$$

- (iii) *The following fixed type of evaluator is accurate: determine input signs and the ordering of their magnitudes exactly; compute the consecutive gaps; then evaluate the collected monomial expansion of $p \circ L$ on the selected chamber.*

Consequently, there is a Turing algorithm that always halts and decides AA-01.

A common C may be used in (ii), since the family of charts is finite. The criterion is global: it includes all zero strata, arbitrarily different scales among gaps, and behavior at infinity. It does not merely test the real zero set.

2 Sufficiency and the decision procedure

2.1 A constant-free canonical evaluator

The sign of a raw input x_i can be found by comparing x_i with its exact negation $-x_i$. Thus no constant zero is needed even for the sign tests. Its magnitude is then an available raw input or its exact negation. Exact comparisons sort these magnitudes; ties are settled by a fixed convention. For fixed n this is a finite comparison tree.

Keep the smallest magnitude as $\hat{y}_1 = y_1$. For $i > 1$, subtract the two *original magnitudes*, not previously computed gaps. Then

$$\hat{y}_i = y_i(1 + \epsilon_i), \quad |\epsilon_i| \leq u. \quad (6)$$

Zero gaps are computed as zero. For $u < 1$ all computed gaps are nonnegative.

On a selected chamber, expand $q = p \circ L$ over the integers before compiling the real-arithmetic evaluator. Because L is linear and $p(0) = 0$, every nonzero monomial has positive degree. Its

computation therefore starts from an available gap rather than a constant one. Use multiplications to form powers and monomials, repeated additions to form positive integer multiples, exact negation for negative coefficients, and additions for the final sum. All repetitions have a fixed, finite length. For $p = 0$, use $x_1 - x_1$.

For clarity, one may implement each positive integer multiple by adding separate occurrences of its monomial value; reusing that value is allowed. Unrolling this fixed evaluator for purposes of error analysis does not assign new independent errors to reused nodes: their existing errors simply occur more than once in the resulting symbolic expression.

Every signed monomial contribution has its exact coefficient multiplied by a finite product of factors $1 + \delta_j$, possibly with repeated factors. Let $R \geq 1$ bound the number of factors in any such product, including the gap errors. It is finite and computable from the constructed evaluator. A coefficient represented by several additions is a positive weighted average of such products after division by its positive integer coefficient. Thus, for $0 < u \leq 1/(2R)$, the relative perturbation of every monomial coefficient is at most

$$E(u) = (1 + u)^R - 1 \leq e^{Ru} - 1 \leq 2Ru. \quad (7)$$

The same bound also controls $1 - (1 - u)^R$. Hence

$$|\tilde{q}(y, \delta) - q(y)| \leq 2Ru q^\#(y) \leq 2CRu |q(y)|. \quad (8)$$

Choose maxima of C and R over the finitely many chambers. Any $0 < u \leq \eta/(2CR)$ gives (1); at zero errors the evaluator returns p exactly. This proves (ii) $\Rightarrow$ (iii), and (iii) $\Rightarrow$ (i) is immediate.

2.2 Finite real-algebraic decision

For each chart the criterion is equivalent to the first-order sentence

$$\exists C \forall y_1 \cdots \forall y_n : \quad C \geq 1 \wedge \left[\left(\bigwedge_{i=1}^n y_i \geq 0 \right) \implies q^\#(y)^2 \leq C^2 q(y)^2 \right]. \quad (9)$$

One may place $C \geq 1$ outside the universal quantifiers instead; this is the same sentence. All coefficients are integers. Squaring introduces no ambiguity because C and $q^\#(y)$ are nonnegative on the domain.

The decision algorithm expands the $2^n n!$ polynomials q , forms their coefficient majorants, and decides each sentence (9) by a complete real quantifier-elimination procedure. Such procedures are effective and terminating; cylindrical algebraic decomposition is one standard choice.⁴ Return YES exactly when every sentence is true. No enumeration over unknown computation trees is needed. No useful running-time bound is asserted.

When all answers are YES, a usable integer C can also be obtained: test the universal conditions with $C = 1, 2, \dots$ until all succeed. This subsequent search halts because a real bound can be increased to an integer. The canonical evaluator and the precision bound above can therefore be supplied as witnesses in addition to the decision.

It remains to prove necessity; that is the substantive part of the argument.

⁴D. S. Arnon, G. E. Collins and S. McCallum, *Cylindrical algebraic decomposition I: The basic algorithm*, SIAM Journal on Computing 13(4) (1984), 865–877; <https://doi.org/10.1137/0213054>. See also Collins’s 1975 quantifier-elimination procedure cited there.

3 Separated executions and replay

3.1 Root provenance and separation

Fix any finite program in the stated model, not yet assumed accurate. During one execution, every available real quantity is an exact signed alias of one of two kinds of *roots*: an original input, or the output of a rounded arithmetic node. Assignment and reuse do not create a new root. Exact negation changes its alias sign, not its root identity. This normalization is valid in the presence of branches: provenance is followed along the selected path. Put

$$M = n + N(P), \quad 0 < u \leq \frac{1}{4}, \quad \gamma = \frac{u}{64M}, \quad \varepsilon = \frac{u\gamma}{16} = \frac{u^2}{1024M}. \quad (10)$$

A bound on the number of rounded nodes on any one path would also suffice in place of $N(P)$.

Lemma 3.1 (A uniformly separated execution). *For each fixed input x , an execution can be chosen with every used error in $[-u/2, u/2]$ such that the following holds. For any two distinct roots with values v, w , at least one of which is a rounded root, either both values are zero or*

$$|v - w| \geq \gamma \max(|v|, |w|), \quad |v + w| \geq \gamma \max(|v|, |w|). \quad (11)$$

There is no separation requirement between two raw input roots.

Proof. Choose errors in execution order. Just before a rounded node, its exact pre-rounding result b and all earlier root values have been determined. Its new value is $v = sb$, where s may be chosen freely in $I = [1 - u/2, 1 + u/2]$, an interval of length u .

If $b = 0$, then $v = 0$. Against a nonzero earlier root both inequalities hold with separation ratio one; a pair of zero values is exempt. Suppose $b \neq 0$. For one earlier value w , first exclude the choices for which

$$|sb - w| < \gamma \max(|sb|, |w|).$$

Since $s > 0$, no choice is excluded if $r = w/b \leq 0$. Otherwise the excluded choices belong to

$$((1 - \gamma)r, r/(1 - \gamma)).$$

If this interval meets $I \subset [1/2, 3/2]$, then $r \leq (3/2)/(1 - \gamma) \leq 2$, using $\gamma \leq 1/4$. Its length is at most

$$r\gamma \frac{2 - \gamma}{1 - \gamma} \leq \frac{16}{3}\gamma < 6\gamma.$$

Replacing w by $-w$ bounds the choices excluded by the other inequality by a further 6γ . There are at most M earlier roots. The total excluded length is less than or equal to $12M\gamma = 3u/16 < u$. A valid $s \in I$ therefore exists. The resulting root is separated from every earlier root, while relations among earlier roots are unchanged.

Continue along the branches selected by these values. Finiteness ensures termination. The choices made along the execution form a legal error vector; fill unused coordinates arbitrarily within their bounds. Adaptive selection in this proof imposes no restriction beyond the model's allowed error box. $\square$

3.2 Primitive forms under gap perturbations

Lemma 3.2 (Relative preservation of primitive forms). *Fix a chart L , $y \geq 0$, and $t_i \in [1 - \varepsilon, 1 + \varepsilon]$, where $0 < \varepsilon < 1$. Set*

$$x = Ly, \quad x' = L \operatorname{diag}(t_1, \dots, t_n)y.$$

For every form ℓ equal to x_i or $x_i \pm x_j$,

$$|\ell(x') - \ell(x)| \leq \varepsilon |\ell(x)|. \quad (12)$$

Consequently, its sign and its zero status are preserved. The same conclusion holds for any exact signed version of these forms.

Proof. Under (2), a raw input is a signed prefix sum of gaps. A sum or difference of two signed prefix sums is either a signed sum of those prefixes or a signed difference of nested prefixes. Therefore $\ell(Ly)$ is zero or has the form $\tau \sum_k a_k y_k$, where $\tau \in \{-1, 1\}$ and $a_k \in \{0, 1, 2\}$. Nonnegativity of the gaps gives

$$|\ell(x') - \ell(x)| \leq \sum_k a_k y_k |t_k - 1| \leq \varepsilon \sum_k a_k y_k = \varepsilon |\ell(x)|.$$

This proof also applies at zero gaps and on chamber intersections. $\square$

Lemma 3.3 (Replay with fixed rounded roots). *Use the constants in (10). Let $x = Ly$ and take a separated execution supplied by Lemma 3.1, with output V . For every $t \in [1 - \varepsilon, 1 + \varepsilon]^n$ and $x' = L \text{diag}(t)y$, there is an execution at x' with errors in $[-u, u]$ such that:*

- (a) *every comparison has the same outcome and the same control path is followed;*
- (b) *every rounded root has exactly its original numerical value;*
- (c) *if V' is the final output, then $|V' - V| \leq \varepsilon |V|$.*

Proof. Proceed through the original execution, keeping the rounded roots fixed inductively. Raw input roots change according to Lemma 3.2. Exact signed aliases preserve these properties automatically.

First consider any comparison reached along the path. If both compared roots are raw, its difference is a signed primitive form and its sign, including equality, is preserved. If both are rounded, their signed values have been held fixed. In the mixed case, write the values as a held signed rounded value v and a signed raw value r . Unless both are zero, separation gives

$$|v - r| \geq \gamma \max(|v|, |r|), \quad |r' - r| \leq \varepsilon |r|.$$

Since $\varepsilon < \gamma$, the comparison cannot change. If both are zero, the raw zero is preserved and equality remains true. Thus the next arithmetic node on the old path is also the next node on the new path.

Let b and b' be its old and new exact pre-rounding results. We claim that $b = 0$ implies $b' = 0$, and otherwise

$$|b'/b - 1| \leq \theta := \varepsilon/\gamma = u/16. \quad (13)$$

Here is a complete accounting of the operand cases.

Operand roots	Operation	Bound for nonzero b
Both rounded	Any allowed operation	0
Both raw	Addition or subtraction	ε
Both raw	Multiplication	$2\varepsilon + \varepsilon^2$
One raw, one rounded	Multiplication	ε
One raw, one rounded	Addition or subtraction	ε/γ

The first line follows from the fixed-root induction. The second is Lemma 3.2, including repeated or oppositely signed aliases. For products, zero factors remain zero; otherwise each varying raw factor changes relatively by at most ε . In the last line, if not both operands are zero, separation

bounds the nonzero sum or difference below by γ times their maximum magnitude, whereas its change is at most ε times that magnitude. A zero sum or difference in this mixed case can only arise from two zero operands, and is preserved. Since $\gamma < 1/3$ and $\varepsilon < 1$, every displayed bound is at most ε/γ . These observations prove (13) and the zero assertion. They also show that a nonzero b cannot become zero.

Let $v = sb$ be the old rounded output, with $|s - 1| \leq u/2$. If $b = b' = 0$, retain s . Otherwise choose

$$s' = \frac{v}{b'} = s \frac{b}{b'}. \quad (14)$$

This makes the new rounded output exactly v . Its error is legal because

$$|s' - 1| \leq \frac{u/2 + \theta}{1 - \theta} \leq \frac{9u}{16 - u} \leq \frac{4u}{7} < u, \quad 0 < u \leq \frac{1}{4}. \quad (15)$$

Thus the induction continues through all arithmetic and comparison nodes. An available final value is a signed rounded root, which is fixed, or a signed raw input, whose relative change is at most ε . This proves (c). $\square$

The lemma does not assume that different occurrences of a reused quantity receive fresh errors. It preserves the stored root itself. Conversely, two separately computed arithmetic outputs are distinct roots, even if their exact zero-error expressions agree; their own multipliers can be chosen separately. Both features are essential to the stated model.

4 Necessity by interpolation

Now suppose that P satisfies (1). Use $\eta = 1/4$ to choose one admissible u , shrinking it to at most $1/4$ if necessary. Fix the resulting M, γ, ε in (10). They do not depend on the input or the selected chart.

For any $x = Ly$ and any $x' = L \operatorname{diag}(t)y$ with $t \in [1 - \varepsilon, 1 + \varepsilon]^n$, combine a separated execution and its replay. Accuracy applies to both error vectors, so

$$|V| \leq \frac{5}{4}|p(x)|, \quad |p(x')| \leq \frac{4}{3}|V'|, \quad |V'| \leq (1 + \varepsilon)|V|.$$

Since $\varepsilon < 1/5$, this gives the uniform local bound

$$|p(L \operatorname{diag}(t)y)| \leq 2|p(Ly)| \quad \text{for every } y \geq 0, \quad t \in [1 - \varepsilon, 1 + \varepsilon]^n. \quad (16)$$

No division by $p(x)$ was used. In particular, the bound remains valid when $p(x) = 0$.

Lemma 4.1 (Coefficient norm from a fixed box). *For fixed integers $n, D \geq 1$ and $\varepsilon > 0$, there is a finite constant $K = K(n, D, \varepsilon)$ such that every polynomial $Q(t) = \sum_{\alpha \in \{0, \dots, D\}^n} b_\alpha t^\alpha$ satisfies*

$$\sum_{\alpha} |b_\alpha| \leq K \max_{t \in [1 - \varepsilon, 1 + \varepsilon]^n} |Q(t)|. \quad (17)$$

Proof. Take the $D + 1$ distinct nodes $r_j = 1 - \varepsilon + 2\varepsilon j/D$, $j = 0, \dots, D$, and the invertible Vandermonde matrix $W = (r_i^j)_{i,j=0}^D$. Values of Q on their tensor product grid equal $(W^{\otimes n}b)$, where b is the vector of coefficients. Thus one may take the sum of the absolute values of all entries of $(W^{-1})^{\otimes n}$ as K . Multiplying the interpolation equations by this inverse and applying the triangle inequality proves (17). $\square$

If $p = 0$, the criterion is immediate. Otherwise set $D = \deg p \geq 1$. Fix a chart and write $q = p \circ L = \sum_{\alpha} c_{\alpha} y^{\alpha}$. For each fixed $y \geq 0$, apply Lemma 4.1 to

$$Q_y(t) = q(y_1 t_1, \dots, y_n t_n) = \sum_{\alpha} c_{\alpha} y^{\alpha} t^{\alpha}.$$

The degree in each variable is at most D , so zero-padding the coefficient array is sufficient. Its coefficient norm is precisely $q^{\#}(y)$, including when some components of y vanish. By (16),

$$q^{\#}(y) \leq K \max_{t \in [1-\varepsilon, 1+\varepsilon]^n} |Q_y(t)| \leq 2K|q(y)| \quad (y \geq 0). \quad (18)$$

Increasing $2K$ to at least one gives (5). This proves (i) $\Rightarrow$ (ii), completing Theorem 1.1.

Corollary 4.2 (A canonical family suffices). *Whenever any accurate tree exists in this model, there is an accurate tree whose only branching determines signs and the ordering of raw input magnitudes. All subsequent arithmetic on the selected branch is a fixed monomial evaluation in its gap coordinates.*

The proof does not require homogeneity, an induction on dominant terms, or an assumption that a leaf reached with nonzero rounding errors computes p when its errors are formally set to zero. In particular, it never substitutes a zero-error identity for the behavior of a branch that need not be reached at zero errors.

Corollary 4.3 (A single uniform tolerance suffices). *Suppose a fixed finite tree satisfies a uniform relative-error guarantee with some one tolerance $0 < \eta_0 < 1$ and some one positive error bound u , on all real inputs and for every allowed error vector. Then the polynomial admits an accurate canonical evaluator in the full sense of (1).*

Proof. Use the same separated execution and replay. Replace the factor two in (16) by the finite factor $(1 + \eta_0)(1 + \varepsilon)/(1 - \eta_0)$. The interpolation argument still establishes (5), and sufficiency supplies arbitrary accuracy. $\square$

5 Face-polynomial obstructions

The coefficient-majorant criterion has an equivalent finite obstruction form. The auxiliary convex-geometric fact below is closely related to the usual face description of the closure of a finite exponential family; no novelty is claimed for that fact.⁵ A proof is included to make the second criterion self-contained.

Let $q(y) = \sum_{\alpha \in S} c_{\alpha} y^{\alpha} \neq 0$, where S is its finite support and each $c_{\alpha} \neq 0$. For a nonempty face F of the convex hull of S , including the whole convex hull, define its face polynomial by

$$q_F(y) = \sum_{\alpha \in S \cap F} c_{\alpha} y^{\alpha}.$$

Proposition 5.1 (Face criterion). *There is a finite constant C with $q^{\#} \leq C|q|$ on $\mathbb{R}_{\geq 0}^n$ if and only if q_F has no zero in $\mathbb{R}_{> 0}^n$ for every nonempty face F .*

Proof. Suppose first that $q_F(z) = 0$ for some $z > 0$. Choose $v \in \mathbb{R}^n$ exposing F as the maximizers of $\alpha \cdot v$ on S , with common maximum b . For the whole face one can take $v = 0$. Along $y_i(s) = z_i \exp(sv_i)$, as $s \rightarrow +\infty$,

$$e^{-sb} q(y(s)) \rightarrow q_F(z) = 0, \quad e^{-sb} q^{\#}(y(s)) \rightarrow (q_F)^{\#}(z) > 0.$$

⁵J. Rauh, T. Kahle and N. Ay, *Support sets in exponential families and oriented matroid theory*, arXiv:0906.5462v2, 16 September 2011, Section 2; <https://arxiv.org/abs/0906.5462v2>.

Thus $|q(y(s))|/q^\#(y(s)) \rightarrow 0$, ruling out domination.

Conversely, failure of domination on the closed orthant implies failure on the positive orthant: a uniform bound on the latter would extend by continuity. Choose $y^{(k)} > 0$ with $|q(y^{(k)})|/q^\#(y^{(k)}) \rightarrow 0$, put $z_k = \log y^{(k)}$ coordinatewise, and consider the positive weights

$$w_\alpha^{(k)} = \frac{|c_\alpha| \exp(\alpha \cdot z_k)}{\sum_{\beta \in S} |c_\beta| \exp(\beta \cdot z_k)}.$$

Pass to a convergent subsequence in the probability simplex, with limit w^* . Its support $T = \{\alpha : w_\alpha^* > 0\}$ is nonempty. Choose $\alpha_0 \in T$, let V be the span of $\alpha - \alpha_0$ for $\alpha \in T$, and decompose $z_k = u_k + v_k$ orthogonally, with $u_k \in V$, $v_k \in V^\perp$. For $\alpha \in T$,

$$(\alpha - \alpha_0) \cdot u_k = \log \frac{w_\alpha^{(k)} |c_{\alpha_0}|}{w_{\alpha_0}^{(k)} |c_\alpha|}$$

converges to a finite limit. A spanning subset of these linear functionals therefore implies $u_k \rightarrow u^* \in V$. For $\beta \notin T$ the same log ratio tends to $-\infty$. Boundedness of u_k gives $(\beta - \alpha_0) \cdot v_k \rightarrow -\infty$. For one sufficiently large k , the vector v_k is consequently constant on T and strictly smaller on every outside support point. It exposes a face with $S \cap F = T$. If $T = S$, take the whole face.

The limits on T have the form

$$w_\alpha^* = \frac{|c_\alpha| \exp(\alpha \cdot u^*)}{\sum_{\beta \in T} |c_\beta| \exp(\beta \cdot u^*)}.$$

Since $\sum_{\alpha \in T} \text{sgn}(c_\alpha) w_\alpha^* = 0$, the positive point $\exp(u^*)$ is a zero of q_F . This proves the converse. $\square$

Corollary 5.2 (Existential obstruction test). *AA-01 has a negative answer exactly when some signed-gap polynomial has a nonempty face polynomial with a strictly positive real zero.*

This also reduces the decision to finitely many existential real sentences, without requiring explicit enumeration of the faces. For a nonzero chart polynomial, quantify a point z , a normal v , a level b , and one selector e_α for each $\alpha \in S$. Impose $z_i > 0$, together with

$$\bigwedge_{\alpha \in S} [(e_\alpha = 1 \wedge \alpha \cdot v = b) \vee (e_\alpha = 0 \wedge \alpha \cdot v < b)],$$

$$\sum_{\alpha \in S} e_\alpha \geq 1, \quad \sum_{\alpha \in S} e_\alpha c_\alpha z^\alpha = 0.$$

The strict inequality for an unselected support point is essential; an arbitrary subset of a face need not itself be a face. The sentence is true exactly when an obstruction exists. The zero polynomial has no obstruction and is treated separately.

A negative certificate can consist of the chart, an integer exposing normal, and a positive real-algebraic zero of the selected face polynomial. The normal can be chosen rational by rational linear feasibility and then scaled to integers; the positive point can be chosen algebraic by real-algebraic sampling. Writing $b = \max_{\alpha \in S} \alpha \cdot v$, the monomial curve $y_i(t) = z_i t^{-v_i}$ has

$$q(y(t)) = t^{-b} Q(t), \quad q^\#(y(t)) = t^{-b} H(t), \quad Q(0) = 0 < H(0),$$

where Q, H are polynomials. Thus its ratio obstruction is directly checkable. The included certificate checker supports rational positive points, not all real-algebraic encodings. The mathematical criterion is not so restricted.

6 Examples and exact certificates

6.1 Difference of squares

For $p(x, y) = x^2 - y^2$, in the chamber $0 \leq x \leq y$ put $x = a$, $y = a + b$. Then $q = -2ab - b^2$ and $q^\# = -q$. The other signed-order chambers similarly have collected coefficients of one sign. Thus $C = 1$ works. The familiar factorization $(x - y)(x + y)$ is also an accurate evaluator in this model.

6.2 A three-term sum

For $p = x + y + z$, choose $x = a$, $y = a + b$, $z = -(a + b + c)$ with $a, b, c \geq 0$. Then $q = a - c$ and $q^\# = a + c$. At the interior gap point $(1, 1, 1)$ the former is zero and the latter is two, so the criterion fails. This agrees with the known obstruction for three-term addition in the independent-error model.²

6.3 A zero-set test is not enough

Consider the nonhomogeneous polynomial

$$p(x, y) = (x^2 - y)^2 + x^2 y^2.$$

It vanishes only at $(0, 0)$: both squares must vanish, so $y = x^2$ and $xy = 0$. In the chamber $0 \leq y \leq x$, put $y = a$, $x = a + b$. Direct expansion gives

$$\begin{aligned} q(a, b) &= 2a^4 + 6a^3b + 7a^2b^2 + 4ab^3 + b^4 \\ &\quad - 2a^3 - 4a^2b - 2ab^2 + a^2. \end{aligned}$$

For $0 < t < 1$, take $a = t^2$ and $b = t - t^2$. Then

$$q(a, b) = t^6, \quad q^\#(a, b) = t^6 + 4t^4, \quad \frac{q^\#(a, b)}{|q(a, b)|} = 1 + \frac{4}{t^2} \longrightarrow \infty.$$

Therefore no accurate finite tree exists, despite the particularly simple real zero set. This example also shows why testing only finite points away from zeros cannot replace the global inequality.

6.4 The Motzkin polynomial

Let

$$p(x, y, z) = x^4 y^2 + x^2 y^4 + z^6 - 3x^2 y^2 z^2.$$

Accurate evaluability of this example with branching is already established in the original work; it is used here as a compatibility check, not as a new result about that polynomial.² Because p is even in all three variables, it suffices to consider their six magnitude orderings. In the four orderings where $|z|$ is smallest or largest, the gap polynomial has nonnegative coefficients, and $C = 1$ works.

In either middle ordering, symmetry between x and y reduces to $x = a$, $z = a + b$, $y = a + b + c$. The resulting polynomial is

$$\begin{aligned} q &= 4a^4 b^2 - 4a^4 bc + 4a^4 c^2 + 12a^3 b^3 - 6a^3 b^2 c + 6a^3 bc^2 + 4a^3 c^3 \\ &\quad + 13a^2 b^4 - 2a^2 b^3 c + 3a^2 b^2 c^2 + 4a^2 bc^3 + a^2 c^4 + 6ab^5 + b^6. \end{aligned}$$

It has three negative coefficients, so sign inspection alone is insufficient. Nevertheless the following exact certificate proves $q^\# \leq 3q$ on $a, b, c \geq 0$:

$$\begin{aligned} 3q - q^\# &= 8a^4(b - c)^2 + 12a^3b(b - c)^2 + 4a^2b^2(b - c)^2 \\ &\quad + 12a^3b^3 + 8a^3c^3 + 22a^2b^4 + 2a^2b^2c^2 \\ &\quad + 8a^2bc^3 + 2a^2c^4 + 12ab^5 + 2b^6. \end{aligned}$$

Every summand is nonnegative. The same bound $C = 3$ therefore works for all 48 signed-order charts. The supplied symbolic test checks all of them, including the 16 signed charts requiring this certificate.

7 Model boundaries and audit points

The proof uses the following specific features rather than a stronger arithmetic model.

Raw inputs and primitive differences. Only input roots are changed in a replay; rounded roots are fixed. Consecutive magnitude gaps preserve all raw forms x_i and $x_i \pm x_j$ relatively because these forms have coefficients of one sign in every chart. An arbitrary polynomial change of coordinates would not have this property.

Independent occurrence errors. At each newly computed rounded root the proof chooses a new multiplier. This is allowed even if two arithmetic occurrences have equal operands. Stored reuse is different: an alias is not rerounded and is held fixed. The method neither assumes nor models correctly rounded IEEE arithmetic, error-free transformations, or probabilistic independence.

Exact comparisons and equality cases. No comparison is assumed to compare exact symbolic polynomials for free. It compares available values. Separation handles mixed comparisons; primitive forms handle raw comparisons. Zeros, repeated operands, exact signed aliases, and tied magnitudes are included explicitly, not discarded as exceptional.

Finiteness and whole-space uniformity. A finite bound on the number of roots is essential to obtain a positive uniform γ . Accuracy must hold for all legal error choices at all real inputs. Restricted domains need not be closed under gap perturbations, and unbounded adaptive loops do not supply this finite root bound. Neither case is covered by the theorem.

No extra exact operations or constants. The argument is for binary $+$, $-$, $\times$ with relative errors and exact negation. No statement is made here for division, fused multiply-add, exact nontrivial scalings, arbitrary polynomial black boxes, or added nonzero constant inputs. The condition $p(0) = 0$ ensures that the constructive direction needs no constant monomial. If an encoding permits zero input variables, $p = 0$ is the only coefficient list satisfying that condition; with the literal requirement that an output be an available real quantity and no input nodes, there is no output-producing tree in that degenerate case. It can be decided separately without affecting the result for $n \geq 1$.

8 Reproducibility and verification status

The companion package contains five complementary families of checks. The separation/replay harness uses exact rational arithmetic throughout, with signed root aliases and conditional operations. Its recorded run covers 100 programs, 2,200 separated executions, and 10,208 replays. All 74,008 replayed rounded-node values were preserved exactly; all 46,464 comparison outcomes checked during replay agreed. Cases include zero pre-results, nonzero cancellation, reuse, exact negation, raw outputs, and equality branches. These finite tests do not prove the general lemmas.

A second control-flow harness allows consecutive comparisons, genuinely variable numbers of arithmetic steps, and early returns. Its 70 finite trees produced 1,680 separated executions and 6,096 replays, preserving 12,771 arithmetic values and 21,343 comparisons. It includes 44 trees whose selected arithmetic path lengths varied and a comparison-only path of length 150.

The symbolic checks verify primitive-form coefficient signs through dimension four, the interpolation identities for the recorded finite examples, the multiplier-budget identity, the Motzkin certificates, the three-term-sum obstruction, and the isolated-zero asymptotic calculation. No floating-point tolerance is used for these checks.

The constant-free compiler is checked on 148 branches from ten example polynomials. All 11,530 arithmetic instructions reference earlier roots and use only the permitted operations and signed aliases. Symbolic interpretation at zero errors recovers the original polynomial identically on every branch. There are also 6,240 exact-rational rounded evaluations checking the bound $|\tilde{q} - q| \leq 2Ruq^\#$; 3,424 of these use a proved coefficient bound and verify relative error at most $1/4$, with exact zero outputs whenever the target vanishes. The Motzkin compiler has global factor bound $R = 36$, so together with $C = 3$ it gives the explicit sufficient precision choice $u \leq \eta/216$. No claim of optimal program size is made.

The face-certificate checks cover two supplied negative examples and 72 constructed rational certificates, split equally among full-face zeros, curves approaching the origin, and curves escaping to infinity. They check 144 exact curve identities and reject 134 false or malformed certificates. These tests also guard against mistaking a nonfacial support subset for a face.

The condition generator accepts an integer coefficient list and emits all chamber polynomials, their majorants, and real quantified sentences. It also emits a Wolfram Language script using `Resolve[... , Reals]`.⁶ A second generator emits the existential obstruction formulas of Section 5. Both generated drivers require literal Boolean backend results and reject unresolved symbolic output. An external general quantifier-elimination backend was *not* executed in the recorded environment; the code generator and the explicit symbolic certificates were executed. The mathematical termination statement uses the standard decision procedure for real closed fields, not an empirical claim that large generated instances are computationally practical.

The proposed resolution rests on the complete mathematical argument in Sections 2–4, with the equivalent obstruction test in Section 5. The tests are diagnostic support, not a formal proof certificate or a substitute for independent mathematical review.

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⁶ Wolfram Research, *Resolve—Wolfram Language Documentation*, <https://reference.wolfram.com/language/ref/Resolve.html>, checked 11 September 2026. The documentation specifies quantifier elimination for polynomial equations and inequalities over the reals.

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